VAYN shoulder actuator — milling quotation

Pack A. Geometry revision 1.4, sheets redrawn 31 October 2026 for the measured 27.5 mm motor. Material revision 30 October 2026. Quantity: one actuator. Units: millimetres.

Quote the parts in section 1 only. The price should include the metal, machining, wire-EDM of the two internal gears, and one carburizing batch on the toothed parts. Do not include nitriding. Do not include anodize. Do not include gear grinding unless a tooth misses its span or pin measurement. Deliver finished parts. Section 4 is not part of the machining price.

The STEP file is the geometric master. Each drawing controls the features printed on that sheet. Sizes with no tolerance on the sheet follow ISO 2768-1 class m and ISO 2768-2 class K.

Part 08, the rear mount plate, is 3D-printed for this build and is not in the STEP. The housing is still machined with the Ø94.5 H7 bore and the eight radial screw holes, so a metal plate can replace the print later.

Two metals

6082-T6 for the housing family: lid, housing, front cover. 6061-T6 may replace 6082 if that is what the shop has; use one aluminium only. 20MnCr5, annealed, for every steel part. Carburize only the teeth: sun-1, both planet sets, ring-1, ring-2, and the sun on carrier-1 front. Mask the carrier-1 front plate. Leave carrier-1 rear and both carrier-2 halves annealed. 16MnCr5 may replace 20MnCr5. Do not use 7075, EN19, EN24, or a second steel.

1. Parts to machine

Part	Qty	Finished size	Mass	Raw stock	Process and finish	Controlled features
Housing family — 6082-T6						
01 Encoder lid	1	Ø59.8 × 2 mm	11.31 g	5 mm 6082-T6 plate, shared with the cover	CNC mill. No anodize	4 × M2 countersunk on Ø54 mm. Ø32 centre opening. The encoder screws are in the rear plate. Cable slot at 0°. Notch at 180° for the phase-lead slot. Screws are a light joint, 0.3 N·m.
15 Housing	1	Ø100 × 90.4 mm, from Z 2 to Z 92.4	349.34 g	Ø105 × 100 mm 6082-T6 bar	CNC lathe and mill, 4th axis for the radial holes. No anodize. Leave the fits and the threads bare.	Rear bore Ø94.5, open through the rear face. Plate plug seats from Z 2 to Z 18.4. 8 × Ø3.2 radial holes with 90° countersinks at Z 6.1, centres 0.1 mm forward of the plug threads. Rear-face notches 4 × 1 mm at 0° and 15 × 3 mm at 180°. Partition 2.5 mm, Z 48.4 to Z 50.9. Labyrinth Ø41.0 H8. Ring-1 bore Ø85 K6. Cartridge bore Ø89 G7. 12 cooling slots, 3 × 20 mm. Two ring-1 dowel slots, width 3.02–3.04 mm, 1.5 mm deep. 8 × M3 in the front face, 6 mm thread engagement.
45 Front cover	1	Ø100 / Ø80 × 3 mm	22.24 g	5 mm 6082-T6 plate, 110 × 110 mm, shared with the lid	CNC mill. No anodize	8 × M2.5 countersunk on Ø93.2 mm. Lip touches the front bearing outer ring only. Hall-sensor pocket at 270°, with a wire slot to the outside edge.
Gears — 20MnCr5						
18 Sun-1	1	Flange Ø40 × 3 mm, teeth Ø16 × 10.4 mm, overall 18.2 mm	42.38 g	Ø45 × 30 mm 20MnCr5, annealed, same heat as the other steel	Turn, hob or shape. Carburize, case 0.12–0.18 mm, 58–62 HRC, core 30–40 HRC. Do not grind the teeth.	18 teeth, module 0.8, 20° pressure angle, profile shift 0. Span over 3 teeth W3 = 6.082 mm, minus 0.015 / plus 0. ISO 1328 grade 7–8 is accepted. Do not grind the teeth. Backlash 0.05 ± 0.02 mm per mesh. 4 × M3 countersunk on Ø31 mm, and 4 × M3 countersunk on Ø25 mm at 45°. Use the pattern the rotor has. Bore Ø3.5 × 10.5 mm for the motor stub. Hub relief Ø18 × 0.5 mm. Flange Ø40 h8.
22 Carrier-1 front and sun-2	1	Plate Ø60 mm, sun Ø20 × 10 mm face, overall about 24 mm	92.03 g	Ø65 × 35 mm 20MnCr5, annealed, same heat	Turn, hob, drill. Carburize the sun only. Case 0.15–0.20 mm, 58–62 HRC. Mask the plate.	Sun-2: 18 teeth, module 1.0. Span over 3 teeth W3 = 7.609 mm, minus 0.015 / plus 0. ISO 1328 grade 7–8 is accepted. Do not grind the teeth. 3 × Ø8 mm R7 holes for the planet pins. 3 × M3 countersunk on Ø44 mm. Rear recess Ø19 × 1.2 mm, so the sun-1 tips clear by 1.5 mm.
24 Planet-1	3	Ø30.4 × 10 mm face, bore Ø12 mm	41.61 g each	Ø35 × 120 mm 20MnCr5 bar, same heat. Hob the three blanks as one stack.	Turn, hob, part off. Carburize, case 0.12–0.18 mm, 58–62 HRC. Grind the bore after hardening.	36 teeth, module 0.8. Span over 5 teeth W5 = 11.008 mm, minus 0.015 / plus 0. ISO 1328 grade 7–8 is accepted. Do not grind the teeth. Bore Ø12 N6, roundness IT5/2, Ra 0.8 µm or finer. Three identical parts. In the STEP they are 24.1, 24.2 and 24.3.
36 Planet-2	3	Ø29 × 10 mm face, bore Ø12 mm	35.24 g each	Same Ø35 bar as planet-1	Turn, hob, part off. Carburize, case 0.15–0.20 mm, 58–62 HRC. Grind the bore after hardening.	27 teeth, module 1.0. Span over 4 teeth W4 = 10.687 mm, minus 0.015 / plus 0. ISO 1328 grade 7–8 is accepted. Do not grind the teeth. This is the high-load mesh, about 1,350 MPa at 30 N·m. Bore Ø12 N6. Three identical parts. In the STEP they are 36.1, 36.2 and 36.3.
16 Ring-1	1	Ø85 h5 × 15.35 mm	190.80 g	Ø105 mm 20MnCr5 bar, annealed, same heat as the other steel	Turn, then wire-EDM the teeth while the steel is still soft. Carburize the teeth only. Case 0.12–0.18 mm, 58–62 HRC. Mask the outside diameter. No tooth grind. Allow 0.03–0.05 mm growth. Finish Ø85 h5 after heat treatment.	Internal gear, 90 teeth, module 0.8. Measure between Ø1.2 mm pins, Mi = 70.813 mm, plus 0.015 / minus 0. 2 × Ø3 mm H7 dowel holes, 7 mm deep from the rear face, on Ø79.5 mm.
Carriers — 20MnCr5						
21 Carrier-1 rear	1	Ø60 × 14 mm, three posts	76.06 g	Same 20MnCr5 bar	CNC mill. Leave annealed. Do not carburize.	3 × Ø8 mm R7 pin holes on Ø43.2 mm. Posts tapped M3, on Ø44 mm. R7 on an h6 pin is 4–28 µm interference.
32 Carrier-2 rear	1	Hub Ø65 × 10 mm, envelope about Ø67 × 21 mm	232.26 g	Same 20MnCr5 bar	CNC mill. Leave annealed. Do not carburize.	Hub Ø65 j5, rear inner-ring seat. 3 × Ø8 mm R7 holes on Ø45 mm, 6 mm deep, ending on a 1.0 mm shoulder. Ø3 mm hole through each shoulder, for venting and punch-out. Rear lightening groove Ø42–Ø59 mm, 3 mm deep. It must not break the pin shoulder. Posts tapped M4. Post faces are the inner-ring shoulders.
Gears — 20MnCr5						
33 Bearing cartridge and ring-2	1	Flange Ø100 × 7.5 mm, spigot Ø89 mm, overall 31.5 mm	350.87 g	Same 20MnCr5 bar	Turn, wire-EDM the ring-2 teeth while the steel is still soft, drill and tap. Carburize the ring-2 teeth only. Case 0.15–0.20 mm, 58–62 HRC. Tap the 8 × M2.5 first and mask them. Mask the Ø89 spigot and the Ø85 bore. No tooth grind. Finish Ø89 and Ø85 after heat treatment.	Spigot Ø89 h6. Bearing bore Ø85 is selective: 3–10 µm larger than the measured 61813 outer diameter. The rear ring must still slide. Ring-2: 72 teeth, module 1.0. Between Ø1.5 mm pins, Mi = 70.499 mm, plus 0.015 / minus 0. 8 × M3 counterbored holes on Ø93 mm. 2 × Ø4 mm dowel holes. 8 × M2.5 tapped for the cover, before carburizing, and masked.
Carriers — 20MnCr5						
42 Carrier-2 front	1	Flange Ø78 mm, hub Ø65 mm, overall 14 mm	313.67 g	Same 20MnCr5 bar	CNC lathe and mill. Leave annealed. Do not carburize. The M5 arm threads are in this steel.	Hub Ø65 j5, front inner-ring seat. 6 × M5 tapped, 8 mm deep, on Ø52 mm. Pilot Ø40 H7 × 2 mm. Runout within 0.02 mm TIR. Face flatness within 0.03 mm. 3 × Ø8 mm R7 holes, blind 6.3 mm deep. 3 × M4 counterbores on Ø54 mm. Ø3 mm radial hole in the rim at Z 102.15 for the homing magnet.
Milled parts, total	17	—	1911 g	—	—	Seventeen solids. Thirteen unique parts.

2. Metal to buy for this one actuator

One aluminium covers the housing family. One heat of 20MnCr5 covers every steel part, so one carburizing coupon qualifies the teeth. A grade mark from the stockist is enough for this first article. No 7075, no EN19, no EN24, no nitriding, no anodize.

Material	Buy	Used for
6082-T6 round. 6061-T6 is allowed if 6082 is not in stock	Ø105 × 100 mm	Housing
6082-T6 plate, same alloy as the bar	5 mm, 110 × 110 mm	Front cover and encoder lid
6082-T6 round, same alloy	none	No magnet holder. The magnet is already on the motor
20MnCr5, annealed, one heat. 16MnCr5 is allowed	Ø105 × 160 mm, one bar	Ring-1, cartridge, carrier-1 rear, both carrier-2 halves
20MnCr5, annealed, the same heat	Ø65 × 35, Ø45 × 30, Ø35 × 120 mm	Carrier-1 front and sun-2, sun-1, six planets

3. Fits and gear checks

Feature	Value
Ring-1 outside in the housing	Ø85 K6 / h5
Cartridge spigot in the housing	Ø89 G7 / h6
61813 outer rings in the cartridge	Bore = measured bearing OD +3 / +10 µm
61813 inner rings on carrier-2	Ø65 j5
Sun-1 flange in the partition	Ø41.0 H8 / Ø40 h8
Future plate plug in the housing	Ø94.5 H7 / k6
Planet bore for HK0810	Ø12 N6
Planet pins in the carriers	Ø8 h6 in Ø8 R7, 4–28 µm interference
Output flange pilot	Ø40 H7 × 2 mm, runout ≤ 0.02 mm
Sun-1 span	W3 = 6.082 −0.015 / 0
Planet-1 span	W5 = 11.008 −0.015 / 0
Sun-2 span	W3 = 7.609 −0.015 / 0
Planet-2 span	W4 = 10.687 −0.015 / 0
Ring-1, over Ø1.2 pins	Mi = 70.813 +0.015 / 0
Ring-2, over Ø1.5 pins	Mi = 70.499 +0.015 / 0
Gear quality	ISO 1328 grade 7–8 is accepted. Do not grind the teeth.
Untoleranced sizes	ISO 2768-1 class m, ISO 2768-2 class K

4. Not in the STEP — do not mill these

These complete the actuator. They are bought, cut from ground shaft, cut from shim stock, or 3D-printed. They are not in the milling price.

Item	Qty	Size	Mass	Material or spec	Why it is not milled
3D-printed this order					
08 Rear mount plate	1	Rim Ø100 × 2 mm plus plug Ø94.5 from Z 2 to Z 18.4	130 g	Printed plastic this build. A later metal plate is 6082-T6.	Not milled. The housing is still cut to accept a metal plug later.
Bought shaft, cut to length					
26 Planet pins 1	3	Ø8 h6 × 17 mm	6.6 g each	52100, through-hardened 58–64 HRC, ground, Ra 0.2 µm or finer	Not milled. Pressed into the carrier. Do not use a g6 shaft or an m6 dowel.
38 Planet pins 2	3	Ø8 h6 × 23 mm	9.0 g each	Same 52100 shaft as the stage-1 pins	Grind a 0.2 mm vent flat on the front 6 mm only, clear of the needle track.
Shim or catalogue ring					
20 Carrier-1 rear washer	1	Ø48 / Ø42 × 0.3 mm	1.0 g	Hardened spring steel, flat within 0.02 mm	Laser, wire-cut, or a shim stack. Not a milled part.
30 Carrier-1 front washer	1	Ø40 / Ø24 × 0.3 mm	1.9 g	Same strip	Runs between the two carriers.
27 28 39 40 Planet washers	12	Ø14 / Ø8.2 × 0.3 mm	0.24 g each	DIN 988 8 × 14 × 0.3, or the same strip	Bore must spin free on the Ø8 h6 pin. Check one sample first.
Shim stock					
48 Motor shims	a 0.25 and 0.5 mm set	Ø50 / Ø21.1 mm, 4 × Ø4.5 on Ø40 at 45°	nominal pack 1.0 mm	Stainless or spring-steel shim stock, flat within 0.02 mm	Nominal pack is 1.0 mm. Not milled.
Catalogue					
12 Motor	1	EaglePower LA8308, Ø92.0 × 27.5 mm flat to flat	336 g	KV90, stub Ø2.9 × 8.9, hex 17.3 across flats, 6.7 proud, Ø10 cover	Purchase.
02 Lid screws	4	M2 × 4 countersunk	0.13 g each	ISO 7046 or 14581, grade 8.8 or A2	Purchase. Torque 0.3 N·m.

Item	Qty	Size	Mass	Material or spec	Why it is not milled
03 Encoder PCB	1	30 × 30 × 1.6 mm	—	FR4, rhombus diagonals 24 and 19, M3	Purchase.
04 Encoder IC	1	TSSOP-14	—	AS5047P-ATSM	Purchase.
05 PCB screws	4	M3 × 6 pan head, rhombus 24/19	0.46 g each	ISO 7045 or 14583	Purchase. Torque 0.4 N·m.
07 Encoder magnet	0	already under the Ø10 cover	—	Glued on the motor hex	Do not buy a second magnet. Do not pin the covered hole.
09 Plate screws	8	M3 × 8 countersunk	0.55 g each	ISO 10642, grade 12.9	Purchase. Used when the metal plate is fitted. Torque 1.2 N·m.
11 Stator screws	4	M4 × 16 countersunk	—	ISO 10642	Purchase. 2.0 N·m maximum into the motor.
17 Ring-1 dowels	2	Ø3 × 8 mm m6	0.44 g each	ISO 8734, hardened	Purchase. Pressed into ring-1, sliding in the housing slot.
19 Sun-1 screws	4	M3 × 5 countersunk	0.38 g each	ISO 10642, grade 12.9	Purchase. Torque 0.8 N·m.
23 Carrier-1 screws	3	M3 × 10 countersunk	0.66 g each	ISO 10642, grade 12.9	Purchase. Torque 1.2 N·m.
25 and 37 Needle bearings	6	HK0810, 8 × 12 × 10 mm	3.1 g catalogue	INA or SKF, open ends	Purchase. Press into the Ø12 N6 bores.
29 Wave spring	1	Smalley SSB-0335	8.6 g in the model	Carbon steel. Bore clears 71.4, outside fits inside 85.	Purchase.
31 Rear output bearing	1	61813-2Z/C3, 65 × 85 × 10 mm	130 g catalogue	Metal shields. C3 is required.	Purchase. Spare bearing is useful for matching the cartridge bore.
34 Cartridge screws	8	M3 × 10 socket head	1.01 g each	ISO 4762, grade 12.9	Purchase. Torque 1.2 N·m.
35 Cartridge dowels	2	Ø4 × 8 mm m6	0.78 g each	ISO 8734, hardened	Purchase.
41 Front output bearing	1	61813-2RS1/C3, 65 × 85 × 10 mm	131 g catalogue	Contact seals. C3 is required.	Purchase. The seals are the gearbox seal.
43 Carrier-2 screws	3	M4 × 16 socket head	2.65 g each	ISO 4762, grade 12.9	Purchase. Torque 4.0 N·m exactly.
44 Homing magnet	1	Ø3 × 2 mm, axial, N52	0.10 g	NdFeB	Purchase. Glued into the output-flange rim.
46 Hall sensor	1	SOT-23	—	TI DRV5033FAQDBZR	Purchase.
47 Cover screws	8	M2.5 × 6 countersunk	0.29 g each	ISO 7046 or 14581	Purchase. Torque 0.8 N·m, low-strength threadlocker.

5. Still open

The magnet is already glued under the Ø10 cover. Do not machine a holder and do not turn a pin into the covered hole. The cover wall thickness is still unmeasured. The cartridge bore Ø85 is machined after the two 61813 bearings are measured.